

TECHNICAL SUPPLEMENT TO NEURAL BELLMAN OPERATORS

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This supplement specifies independent algebraic and trajectory checks, the query implementation and reliability contracts, and all individual process observations used in the revised article. Exact retained editions contain the preceding whole-domain proof, numerical evidence and broader applications without deletion.

1. INDEPENDENT ALGEBRAIC AND NUMERICAL CHECKS

The article contains complete proofs of the parabolic minorant, recursive query, gated refinement and certificate-localization statements. Deterministic tests supplement those arguments. They do not substitute a finite collection of examples for a proof over all states. The retained query suite has 24 tests, the gated suite 17, and the new independent suite 22. They include exact rational identities, interval boundary cases, inaccurate predictors, precision failure, continuous quadrature checks and deliberately corrupted records.

1.1. *Curvature and regularity*

The state curvature slack in the original scalar model is

$$\frac{4}{d} - \frac{15}{16} \frac{27}{8d} = \frac{107}{128d} > 0.$$

The independently written recurrence starts at $dG_0 = 10$, $dM_0 = 26$ and uses

$$dG_r = \frac{1037}{128} + \frac{705}{1024} dG_{r-1}, \quad dM_r = 22 + \frac{21}{256} + \frac{15}{16} \left(\frac{9}{16} dM_{r-1} + \frac{1}{8} dG_{r-1} \right).$$

It agrees exactly with the implemented rational coefficients. The induction bounds $dG_r \leq 27$ and $dM_r \leq 54$ imply the action-curvature, derivative and quadrature coefficients printed in the article. The terminal action coefficients $41/32$ and $9/4$ are

separately derived from the alternating action exposures and checked against the original rational objective and derivative.

For a uniform innovation on $[-1/32, 1/32]$, let $z_j = (2j + 1 - q)/(32q)$ be the conditional means of q equal bins. The exact identity

$$\frac{1}{3072} - \frac{1}{q} \sum_{j=0}^{q-1} z_j^2 = \frac{1}{3072q^2}$$

gives the conditional variance discarded by midpoint evaluation. Convexity supplies the lower inequality and semiconcavity supplies the upper remainder. The fitted model supplies neither. Additional primitive floating uncertainty is expanded with the proven Lipschitz constant before a child interval is accepted.

1.2. Feasibility at boundaries

The capacity $c(x) = 1/8 + \sum_j x_j/(8d)$ is affine and nondecreasing. Barycentric combinations of feasible nodal controls are feasible at the interpolated state because their summed controls are bounded by the same barycentric combination of capacities. The tests include vertices, faces, equality at capacity and two-coordinate controls. Downward state acquisition preserves the ordering of capacities. The implemented query action is rounded downward on the thirty-two-bit lattice before its original-law endpoint is evaluated. The numerical certificate therefore belongs to the action actually used.

For a truncated capacity endpoint $\underline{c}$, the omitted interval is charged by $\Lambda(c - \underline{c})$. No lower envelope is permitted to ignore that interval. Projected derivative certification at zero and at capacity uses the appropriate one-sided optimality sign; a constrained optimum need not have derivative zero. The example $f(a) = a$ shows why a boundary prediction-error term can be first order even though an interior one is second order.

1.3. Localization and corrupt-record tests

The new rational tests compute exact parabolic lower bounds for linear and quadratic objectives, including both boundary minimizers and an interior minimizer. They check the certificate-localization bound with endpoint uncertainty, a

shortened capacity interval, all three parabolic-minimum cases and the running-bound identity. A constant objective with an exact optimal witness and a deliberately loose curvature allowance provides a useful counterexample: prediction accuracy alone does not close a coarse lower certificate.

The trajectory checker is tested by separately corrupting an acquired state, a returned action, a reported local gap, an exact path cost and the requested target. Each corrupted object is rejected. This verifies that the replay is checking economic implementation and accuracy fields rather than only accepting a success flag. It is not a claim that these five mutations exhaust all possible implementation faults.

2. FINITE-PRECISION AND WORKLOAD CONTRACTS

A service receives the dimension, horizon, target, workload size, mode, fitting seed and repetition number. Those fields are bound to its summary and parent receipt. Every scientific source freeze is checked before replay, including the imported original primitives. Each returned trace contains true initial and innovation dyadic indices, acquired states, actions, original-law lower and upper endpoints, local gaps and probe counts. Models, exact costs, traces, summaries and clocks have separate SHA-256 identities.

The local request is the recorded binary64 rounding of $\epsilon / (2 \sum_{t < T} \beta^t)$. The global policy allowance uses this actual local request and a rational acquisition expansion before upward conversion. For every declared target and horizon, the independent suite checks the outward direction and the final comparison with ϵ . It also checks the recovery condition $\delta_r > 8\Lambda_r 2^{-32}$ at the smallest descendant request. A fixed-precision failure cannot be reclassified as a valid certificate because another workload previously returned.

The finite return theorem separates real-arithmetic refinement from executable recovery. In the floating implementation, a failed width guard or refinement cap invokes exact-rational enumeration, with separately counted queries. That procedure refines only the required descendants and never loads a conventional full-state policy. Its rational bit complexity can differ substantially from the floating operation count. No recovery was observed in the declared workloads; the manuscript does not infer that recovery is costless or unnecessary in general.

2.1. Independent exact trajectories

For rational x , a and z , the replay computes the transition as

$$F_j(x, a, z) = \frac{1}{16} + \frac{9}{16}x_j + \frac{1}{8}x_{j+1} - \frac{1}{16}x_jx_{j+1} + B_ja + s_jz.$$

This is coded independently of the original interval routine. The stage and terminal costs are likewise evaluated from their polynomial and positive-part definitions. The true state remains rational across dates; only the observation and selected action are quantized. The replay checks every acquired state, true-state capacity, invariant-domain condition and complete discounted path cost exactly. It does not replace the evolving true state with its acquired version when computing the economic loss.

Across both frozen catalogues, 55,296 decision records are bound to valid traces. The repeated models and traces match exactly across the two repetitions. Consequently the numerical re-query and rational reconstruction need only be executed once for each byte-identical trace/model combination; each repeated receipt, request and identity is still checked. There are 27,648 distinct recorded decision observations re-queried. This economy of verification does not change the number of original process measurements or create new independent observations.

3. COMPLETE INDIVIDUAL SERVICE OBSERVATIONS

The following tables retain every declared service. A denotes adaptive parabolic search, B bisection, RI unconditional ReLU insertion, RG gated ReLU routing, QI unconditional quadratic insertion and QG gated quadratic routing. The first catalogue has no routing arm. Seed zero denotes a conventional method with no fit. Repetition zero and repetition one are two distinct isolated-process measurements; identical numerical traces do not imply identical clocks.

“Train Q” is the number of Bellman action queries paid to construct training labels. “Deploy Q” includes the actual policy’s recursive queries and its terminal kernel. These counts are algorithmic operations, not statistically independent sample sizes. “Warn.” is the number of retained fitting warnings at service dates. Seconds are complete parent clocks, rounded for display; exact stored binary64

values determine the derived accounts. Individual peak memory, maximum live batch, fitted parameters, path costs and receipt hashes remain in the accompanying machine-readable audit.

TABLE I: R63 complete individual service observations

d	T	Target	Method	Seed	Rep.	Seconds	Train Q	Deploy Q	Warn.
2	2	1/32	A	0	0	0.2657	0	1280	0
2	2	1/32	A	0	1	0.2583	0	1280	0
2	2	1/32	B	0	0	0.2578	0	1295	0
2	2	1/32	B	0	1	0.2561	0	1295	0
2	2	1/32	RI	6301	0	0.6489	6912	1469	0
2	2	1/32	RI	6301	1	0.6467	6912	1469	0
2	2	1/32	RI	6302	0	0.6386	6912	1415	0
2	2	1/32	RI	6302	1	0.6366	6912	1415	0
2	2	1/32	QI	6301	0	0.6116	6912	1409	0
2	2	1/32	QI	6301	1	0.6089	6912	1409	0
2	2	1/32	QI	6302	0	0.6091	6912	1391	0
2	2	1/32	QI	6302	1	0.6057	6912	1391	0
2	2	1/128	A	0	0	0.2680	0	2353	0
2	2	1/128	A	0	1	0.2714	0	2353	0
2	2	1/128	B	0	0	0.2798	0	2533	0
2	2	1/128	B	0	1	0.2803	0	2533	0
2	2	1/128	RI	6301	0	0.6664	6912	2598	0
2	2	1/128	RI	6301	1	0.6588	6912	2598	0
2	2	1/128	RI	6302	0	0.6531	6912	2553	0
2	2	1/128	RI	6302	1	0.6580	6912	2553	0
2	2	1/128	QI	6301	0	0.6202	6912	2533	0
2	2	1/128	QI	6301	1	0.6254	6912	2533	0
2	2	1/128	QI	6302	0	1.4713	6912	2468	0
2	2	1/128	QI	6302	1	0.6208	6912	2468	0
8	2	1/32	A	0	0	0.3802	0	1265	0
8	2	1/32	A	0	1	0.3774	0	1265	0
8	2	1/32	B	0	0	0.3555	0	1328	0

	<i>d</i>	<i>T</i>	Target	Method	Seed	Rep.	Seconds	Train Q	Deploy Q	Warn.	
1											1
2	8	2	1/32	B	0	1	0.3577	0	1328	0	2
3	8	2	1/32	RI	6301	0	0.7536	6912	1478	1	3
4	8	2	1/32	RI	6301	1	0.7745	6912	1478	1	4
5	8	2	1/32	RI	6302	0	0.7871	6912	1427	1	5
6	8	2	1/32	RI	6302	1	0.7837	6912	1427	1	6
7	8	2	1/32	QI	6301	0	0.7302	6912	1460	0	7
8	8	2	1/32	QI	6301	1	0.7384	6912	1460	0	8
9	8	2	1/32	QI	6302	0	0.7295	6912	1496	0	9
10	8	2	1/32	QI	6302	1	0.7316	6912	1496	0	10
11	8	2	1/128	A	0	0	0.4069	0	2393	0	11
12	8	2	1/128	A	0	1	0.4141	0	2393	0	12
13	8	2	1/128	B	0	0	0.4279	0	2703	0	13
14	8	2	1/128	B	0	1	0.4259	0	2703	0	14
15	8	2	1/128	RI	6301	0	0.8082	6912	2628	1	15
16	8	2	1/128	RI	6301	1	0.8106	6912	2628	1	16
17	8	2	1/128	RI	6302	0	0.8044	6912	2603	1	17
18	8	2	1/128	RI	6302	1	0.8117	6912	2603	1	18
19	8	2	1/128	QI	6301	0	0.7594	6912	2618	0	19
20	8	2	1/128	QI	6301	1	0.7649	6912	2618	0	20
21	8	2	1/128	QI	6302	0	0.7682	6912	2653	0	21
22	8	2	1/128	QI	6302	1	0.7646	6912	2653	0	22
23	16	2	1/32	A	0	0	0.5348	0	1265	0	23
24	16	2	1/32	A	0	1	0.5357	0	1265	0	24
25	16	2	1/32	B	0	0	0.5045	0	1316	0	25
26	16	2	1/32	B	0	1	0.5002	0	1316	0	26
27	16	2	1/32	RI	6301	0	0.9419	6912	1448	1	27
28	16	2	1/32	RI	6301	1	0.9454	6912	1448	1	28
29	16	2	1/32	RI	6302	0	0.9458	6912	1424	1	29
30	16	2	1/32	RI	6302	1	0.9340	6912	1424	1	30
31	16	2	1/32	QI	6301	0	0.9000	6912	1427	0	31
32	16	2	1/32	QI	6301	1	0.8981	6912	1427	0	32
	16	2	1/32	QI	6302	0	0.9056	6912	1445	0	
	16	2	1/32	QI	6302	1	0.8959	6912	1445	0	

	d	T	Target	Method	Seed	Rep.	Seconds	Train Q	Deploy Q	Warn.	
1											1
2	16	2	1/128	A	0	0	0.5493	0	2278	0	2
3	16	2	1/128	A	0	1	0.5499	0	2278	0	3
4	16	2	1/128	B	0	0	0.6272	0	2583	0	4
5	16	2	1/128	B	0	1	0.6299	0	2583	0	5
6	16	2	1/128	RI	6301	0	1.0003	6912	2643	1	6
7	16	2	1/128	RI	6301	1	0.9952	6912	2643	1	7
8	16	2	1/128	RI	6302	0	0.9967	6912	2538	1	8
9	16	2	1/128	RI	6302	1	0.9965	6912	2538	1	9
10	16	2	1/128	QI	6301	0	0.9228	6912	2538	0	10
11	16	2	1/128	QI	6301	1	0.9151	6912	2538	0	11
12	16	2	1/128	QI	6302	0	0.9219	6912	2573	0	12
13	16	2	1/128	QI	6302	1	0.9137	6912	2573	0	13
14	8	3	1/32	A	0	0	0.9580	0	6638	0	14
15	8	3	1/32	A	0	1	0.9643	0	6638	0	15
16	8	3	1/32	B	0	0	1.0045	0	7376	0	16
17	8	3	1/32	B	0	1	1.0065	0	7376	0	17
18	8	3	1/32	RI	6301	0	1.9047	155904	9402	2	18
19	8	3	1/32	RI	6301	1	1.9126	155904	9402	2	19
20	8	3	1/32	RI	6302	0	1.9265	155904	8042	2	20
21	8	3	1/32	RI	6302	1	1.9255	155904	8042	2	21
22	8	3	1/32	QI	6301	0	1.8335	155904	8404	0	22
23	8	3	1/32	QI	6301	1	1.8324	155904	8404	0	23
24	8	3	1/32	QI	6302	0	1.8350	155904	8754	0	24
25	8	3	1/32	QI	6302	1	1.8372	155904	8754	0	25
26	8	3	1/128	A	0	0	1.4140	0	27165	0	26
27	8	3	1/128	A	0	1	1.4087	0	27165	0	27
28	8	3	1/128	B	0	0	1.8735	0	34394	0	28
29	8	3	1/128	B	0	1	1.8832	0	34394	0	29
30	8	3	1/128	RI	6301	0	2.3955	155904	35526	2	30
31	8	3	1/128	RI	6301	1	2.3911	155904	35526	2	31
32	8	3	1/128	RI	6302	0	2.4166	155904	31210	2	32
	8	3	1/128	RI	6302	1	2.4197	155904	31210	2	
	8	3	1/128	QI	6301	0	2.3370	155904	32253	0	

d	T	Target	Method	Seed	Rep.	Seconds	Train Q	Deploy Q	Warn.
8	3	1/128	QI	6301	1	2.3315	155904	32253	0
8	3	1/128	QI	6302	0	2.3348	155904	33324	0
8	3	1/128	QI	6302	1	2.3232	155904	33324	0

TABLE II: R64 complete individual service observations

d	T	Target	Method	Seed	Rep.	Seconds	Train Q	Deploy Q	Warn.
2	2	1/32	A	0	0	0.2586	0	1298	0
2	2	1/32	A	0	1	0.2542	0	1298	0
2	2	1/32	B	0	0	0.2515	0	1310	0
2	2	1/32	B	0	1	0.2523	0	1310	0
2	2	1/32	RI	6401	0	0.7141	6912	1415	1
2	2	1/32	RI	6401	1	0.6452	6912	1415	1
2	2	1/32	RI	6402	0	0.6510	6912	1439	0
2	2	1/32	RI	6402	1	0.6498	6912	1439	0
2	2	1/32	RG	6401	0	0.6662	6912	1298	1
2	2	1/32	RG	6401	1	0.6555	6912	1298	1
2	2	1/32	RG	6402	0	0.6517	6912	1298	0
2	2	1/32	RG	6402	1	0.6579	6912	1298	0
2	2	1/32	QI	6401	0	0.6250	6912	1418	0
2	2	1/32	QI	6401	1	0.6097	6912	1418	0
2	2	1/32	QI	6402	0	0.6127	6912	1424	0
2	2	1/32	QI	6402	1	0.6144	6912	1424	0
2	2	1/32	QG	6401	0	0.6138	6912	1295	0
2	2	1/32	QG	6401	1	0.9454	6912	1295	0
2	2	1/32	QG	6402	0	0.6151	6912	1295	0
2	2	1/32	QG	6402	1	0.6099	6912	1295	0
2	2	1/128	A	0	0	0.2698	0	2508	0
2	2	1/128	A	0	1	0.2700	0	2508	0
2	2	1/128	B	0	0	0.2805	0	2658	0
2	2	1/128	B	0	1	0.2830	0	2658	0

	d	T	Target	Method	Seed	Rep.	Seconds	Train Q	Deploy Q	Warn.	
1											1
2	2	2	1/128	RI	6401	0	0.6702	6912	2593	1	2
3	2	2	1/128	RI	6401	1	0.6624	6912	2593	1	3
4	2	2	1/128	RI	6402	0	0.6475	6912	2643	0	4
5	2	2	1/128	RI	6402	1	0.6408	6912	2643	0	5
6	2	2	1/128	RG	6401	0	0.6561	6912	2438	1	6
7	2	2	1/128	RG	6401	1	0.6555	6912	2438	1	7
8	2	2	1/128	RG	6402	0	0.6493	6912	2428	0	8
9	2	2	1/128	RG	6402	1	0.6603	6912	2428	0	9
10	2	2	1/128	QI	6401	0	0.6283	6912	2633	0	10
11	2	2	1/128	QI	6401	1	0.6215	6912	2633	0	11
12	2	2	1/128	QI	6402	0	0.6232	6912	2613	0	12
13	2	2	1/128	QI	6402	1	0.6243	6912	2613	0	13
14	2	2	1/128	QG	6401	0	0.6320	6912	2473	0	14
15	2	2	1/128	QG	6401	1	0.6224	6912	2473	0	15
16	2	2	1/128	QG	6402	0	0.6241	6912	2463	0	16
17	2	2	1/128	QG	6402	1	0.6367	6912	2463	0	17
18	8	2	1/32	A	0	0	0.3777	0	1280	0	18
19	8	2	1/32	A	0	1	0.3821	0	1280	0	19
20	8	2	1/32	B	0	0	0.3616	0	1325	0	20
21	8	2	1/32	B	0	1	0.3586	0	1325	0	21
22	8	2	1/32	RI	6401	0	0.7551	6912	1472	1	22
23	8	2	1/32	RI	6401	1	0.7606	6912	1472	1	23
24	8	2	1/32	RI	6402	0	0.7788	6912	1472	1	24
25	8	2	1/32	RI	6402	1	0.7917	6912	1472	1	25
26	8	2	1/32	RG	6401	0	0.7678	6912	1277	1	26
27	8	2	1/32	RG	6401	1	0.7616	6912	1277	1	27
28	8	2	1/32	RG	6402	0	0.7869	6912	1280	1	28
29	8	2	1/32	RG	6402	1	0.7749	6912	1280	1	29
30	8	2	1/32	QI	6401	0	0.7419	6912	1451	0	30
31	8	2	1/32	QI	6401	1	0.7405	6912	1451	0	31
32	8	2	1/32	QI	6402	0	0.7417	6912	1472	0	32
	8	2	1/32	QI	6402	1	0.7400	6912	1472	0	
	8	2	1/32	QG	6401	0	0.7451	6912	1280	0	

	<i>d</i>	<i>T</i>	Target	Method	Seed	Rep.	Seconds	Train Q	Deploy Q	Warn.	
1											1
2	8	2	1/32	QG	6401	1	0.7616	6912	1280	0	2
3	8	2	1/32	QG	6402	0	0.7329	6912	1277	0	3
4	8	2	1/32	QG	6402	1	0.7224	6912	1277	0	4
5	8	2	1/128	A	0	0	0.4154	0	2373	0	5
6	8	2	1/128	A	0	1	0.4165	0	2373	0	6
7	8	2	1/128	B	0	0	0.4437	0	2648	0	7
8	8	2	1/128	B	0	1	0.4373	0	2648	0	8
9	8	2	1/128	RI	6401	0	0.8051	6912	2608	1	9
10	8	2	1/128	RI	6401	1	0.7990	6912	2608	1	10
11	8	2	1/128	RI	6402	0	0.8215	6912	2618	1	11
12	8	2	1/128	RI	6402	1	0.8073	6912	2618	1	12
13	8	2	1/128	RG	6401	0	0.8055	6912	2358	1	13
14	8	2	1/128	RG	6401	1	0.8242	6912	2358	1	14
15	8	2	1/128	RG	6402	0	0.8069	6912	2393	1	15
16	8	2	1/128	RG	6402	1	0.8057	6912	2393	1	16
17	8	2	1/128	QI	6401	0	0.7548	6912	2573	0	17
18	8	2	1/128	QI	6401	1	0.7619	6912	2573	0	18
19	8	2	1/128	QI	6402	0	0.7533	6912	2603	0	19
20	8	2	1/128	QI	6402	1	0.7467	6912	2603	0	20
21	8	2	1/128	QG	6401	0	0.7602	6912	2373	0	21
22	8	2	1/128	QG	6401	1	0.7848	6912	2373	0	22
23	8	2	1/128	QG	6402	0	0.7703	6912	2358	0	23
24	8	2	1/128	QG	6402	1	0.7753	6912	2358	0	24
25	16	2	1/32	A	0	0	0.5383	0	1286	0	25
26	16	2	1/32	A	0	1	0.5341	0	1286	0	26
27	16	2	1/32	B	0	0	0.4973	0	1328	0	27
28	16	2	1/32	B	0	1	0.4955	0	1328	0	28
29	16	2	1/32	RI	6401	0	0.9512	6912	1433	1	29
30	16	2	1/32	RI	6401	1	0.9526	6912	1433	1	30
31	16	2	1/32	RI	6402	0	0.9195	6912	1478	1	31
32	16	2	1/32	RI	6402	1	0.9131	6912	1478	1	32
	16	2	1/32	RG	6401	0	0.9118	6912	1283	1	
	16	2	1/32	RG	6401	1	0.9231	6912	1283	1	

	d	T	Target	Method	Seed	Rep.	Seconds	Train Q	Deploy Q	Warn.	
1											1
2	16	2	1/32	RG	6402	0	0.9097	6912	1280	1	2
3	16	2	1/32	RG	6402	1	0.9074	6912	1280	1	3
4	16	2	1/32	QI	6401	0	0.8947	6912	1451	0	4
5	16	2	1/32	QI	6401	1	0.8840	6912	1451	0	5
6	16	2	1/32	QI	6402	0	0.8905	6912	1463	0	6
7	16	2	1/32	QI	6402	1	0.8912	6912	1463	0	7
8	16	2	1/32	QG	6401	0	0.8558	6912	1283	0	8
9	16	2	1/32	QG	6401	1	0.8530	6912	1283	0	9
10	16	2	1/32	QG	6402	0	0.8644	6912	1283	0	10
11	16	2	1/32	QG	6402	1	0.8568	6912	1283	0	11
12	16	2	1/128	A	0	0	0.5504	0	2318	0	12
13	16	2	1/128	A	0	1	0.5527	0	2318	0	13
14	16	2	1/128	B	0	0	0.6213	0	2633	0	14
15	16	2	1/128	B	0	1	0.6218	0	2633	0	15
16	16	2	1/128	RI	6401	0	0.9672	6912	2608	1	16
17	16	2	1/128	RI	6401	1	0.9621	6912	2608	1	17
18	16	2	1/128	RI	6402	0	0.9975	6912	2548	1	18
19	16	2	1/128	RI	6402	1	0.9898	6912	2548	1	19
20	16	2	1/128	RG	6401	0	0.9855	6912	2368	1	20
21	16	2	1/128	RG	6401	1	0.9849	6912	2368	1	21
22	16	2	1/128	RG	6402	0	0.9890	6912	2348	1	22
23	16	2	1/128	RG	6402	1	1.0064	6912	2348	1	23
24	16	2	1/128	QI	6401	0	0.9144	6912	2598	0	24
25	16	2	1/128	QI	6401	1	0.9251	6912	2598	0	25
26	16	2	1/128	QI	6402	0	0.9089	6912	2608	0	26
27	16	2	1/128	QI	6402	1	0.9108	6912	2608	0	27
28	16	2	1/128	QG	6401	0	0.9393	6912	2373	0	28
29	16	2	1/128	QG	6401	1	0.9502	6912	2373	0	29
30	16	2	1/128	QG	6402	0	0.9316	6912	2348	0	30
31	16	2	1/128	QG	6402	1	0.9021	6912	2348	0	31
32	8	3	1/32	A	0	0	0.9763	0	7042	0	32
	8	3	1/32	A	0	1	0.9923	0	7042	0	
	8	3	1/32	B	0	0	1.0346	0	8112	0	

	<i>d</i>	<i>T</i>	Target	Method	Seed	Rep.	Seconds	Train Q	Deploy Q	Warn.	
1											1
2	8	3	1/32	B	0	1	1.0266	0	8112	0	2
3	8	3	1/32	RI	6401	0	1.9628	155904	9711	2	3
4	8	3	1/32	RI	6401	1	1.9639	155904	9711	2	4
5	8	3	1/32	RI	6402	0	1.9298	155904	8317	2	5
6	8	3	1/32	RI	6402	1	1.9611	155904	8317	2	6
7	8	3	1/32	RG	6401	0	1.8078	155904	6965	2	7
8	8	3	1/32	RG	6401	1	1.8173	155904	6965	2	8
9	8	3	1/32	RG	6402	0	1.8316	155904	6930	2	9
10	8	3	1/32	RG	6402	1	1.7950	155904	6930	2	10
11	8	3	1/32	QI	6401	0	1.8207	155904	8539	0	11
12	8	3	1/32	QI	6401	1	1.8391	155904	8539	0	12
13	8	3	1/32	QI	6402	0	1.8545	155904	8669	0	13
14	8	3	1/32	QI	6402	1	1.8405	155904	8669	0	14
15	8	3	1/32	QG	6401	0	1.8065	155904	7081	0	15
16	8	3	1/32	QG	6401	1	1.7982	155904	7081	0	16
17	8	3	1/32	QG	6402	0	1.7837	155904	6996	0	17
18	8	3	1/32	QG	6402	1	1.8069	155904	6996	0	18
19	8	3	1/128	A	0	0	1.4404	0	31015	0	19
20	8	3	1/128	A	0	1	1.4604	0	31015	0	20
21	8	3	1/128	B	0	0	1.9562	0	39768	0	21
22	8	3	1/128	B	0	1	1.9488	0	39768	0	22
23	8	3	1/128	RI	6401	0	2.4113	155904	39472	2	23
24	8	3	1/128	RI	6401	1	2.4064	155904	39472	2	24
25	8	3	1/128	RI	6402	0	2.4844	155904	35566	2	25
26	8	3	1/128	RI	6402	1	2.4596	155904	35566	2	26
27	8	3	1/128	RG	6401	0	2.3900	155904	30547	2	27
28	8	3	1/128	RG	6401	1	2.3792	155904	30547	2	28
29	8	3	1/128	RG	6402	0	2.5831	155904	31601	2	29
30	8	3	1/128	RG	6402	1	2.5721	155904	31601	2	30
31	8	3	1/128	QI	6401	0	2.3349	155904	34705	0	31
32	8	3	1/128	QI	6401	1	2.3239	155904	34705	0	32
	8	3	1/128	QI	6402	0	2.3191	155904	35886	0	
	8	3	1/128	QI	6402	1	2.3214	155904	35886	0	

d	T	Target	Method	Seed	Rep.	Seconds	Train Q	Deploy Q	Warn.
8	3	1/128	QG	6401	0	2.3186	155904	30908	0
8	3	1/128	QG	6401	1	2.2859	155904	30908	0
8	3	1/128	QG	6402	0	2.2838	155904	30889	0
8	3	1/128	QG	6402	1	2.2779	155904	30889	0

4. STATISTICAL AND ECONOMIC INTERPRETATION

The workload generation rule and named boundary replacements are fixed in each source-frozen protocol. Its dyadic average has an exact reproducible value but is not an independent continuous-law estimate. The earlier bounded-sample cost intervals remain attached to their original frozen policies and their declared simultaneous families (Maurer and Pontil, 2009). The query-service catalogue adds neither new confidence intervals for expected gains nor additional optimizer replications to that earlier study.

A complete cold-start time is a directly observed service cost on the recorded machine. A median of two conventional runs or four learned runs is descriptive. Fixed affinity and single-thread configuration remove some sources of variation but do not control CPU frequency, system interference or another platform. We therefore retain all clocks, logical operation counts and process memory, and make no hardware-uniform timing probability claim.

For a decision maker with a declared dollar conversion per model-loss unit, a price per computation second and a deployment fee, an independently valid gain interval can be converted to a monetary purchase interval as shown in the article. A uniform regret bound instead supports an accuracy constraint. An accuracy constraint is not a measured welfare gain. Likewise, a conditional arithmetic break-even workload count does not establish that an unobserved future distribution will produce the same query savings.

5. PRESERVATION AND REPRODUCIBLE PUBLICATION

The original NBO title, own-future construction, investment primitives and Bellman objective remain in the active article. Chronological repetitions are removed

from its reading path, not from the research record. The exact retained manuscript tree contains every earlier text source, with a location map for every distinct label. The preceding article and supplement and their complete broader companions are compiled separately. Their recursive-utility, diffusion and game results retain their original assumptions and are not asserted as numerical consequences of the scalar expected-cost query theorem.

The offline build runs the two inherited query test suites, the independent tests, all source and record checks, and the table derivation before compiling the active article, supplement and response. The four preserved companions are compiled from their unchanged sources. Undefined references, duplicate labels, missing characters, fatal errors and overfull boxes are publication failures. Any nonfatal class-font substitutions are recorded rather than described as absent. A clean publication rebuild deletes generated tables and PDFs and checks the resulting document text and page counts. Neither compilation nor record identity is external mathematical acceptance.

REFERENCES

MAURER, ANDREAS AND MASSIMILIANO PONTIL (2009): "Empirical Bernstein Bounds and Sample Variance Penalization," in *Proceedings of the 22nd Annual Conference on Learning Theory*, arXiv:0907.3740. [13]